

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Embedded Programming

Guidelines for Preparation and Submission of Embedded Hardware Project Report

Dear Students,

All students are instructed to prepare the **Embedded Hardware Project Report** according to the following prescribed **Table of Contents**. The report must clearly document the design, development, implementation, testing, and results of the hardware project.

Students are requested to strictly follow the given structure while preparing the final project report.

REPORT FORMAT

Abstract and Keywords

- Abstract
- Keywords

Chapter 1 – Introduction

- 1.1 Project Overview
- 1.2 Problem Statement
- 1.3 Objectives
- 1.4 Scope of the Project

Chapter 2 – Literature Survey

- 2.1 Related Existing Projects
- 2.2 Review of Existing Technologies
- 2.3 Comparison of Existing Systems

Chapter 3 – Existing System

- 3.1 Description
- 3.2 Drawbacks and Limitations

Chapter 4 – Proposed System

- 4.1 Proposed Solution
- 4.2 Features
- 4.3 Advantages

Chapter 5 – System Design

- 5.1 Block Diagram
- 5.2 Working Principle

5.3 Circuit Diagram

5.4 Flowchart

Chapter 6 – Hardware Components

6.1 Microcontroller / Development Board

6.2 Sensors

6.3 Actuators

6.4 Display / Output Devices

6.5 Communication Module

6.6 Power Supply

6.7 Other Components

Chapter 7 – Software Implementation

7.1 Software Requirements

7.2 Embedded C Program

7.3 Python Program

7.4 Simulation – Wokwi / Raspberry Pi / Thonny (*if applicable*)

Chapter 8 – Hardware Implementation

8.1 Circuit Assembly

8.2 Hardware Connections

8.3 Prototype Development

8.4 Working of the Hardware Model

8.5 Photographs of the Developed Model

Chapter 9 – Testing and Results

9.1 Testing Procedure

9.2 Test Cases

9.3 Experimental Results

9.4 Output Screenshots

9.5 Performance Analysis

Chapter 10 – Applications

Chapter 11 – Advantages and Limitations

Chapter 12 – Conclusion and Future Scope

Chapter 13 – References

Chapter 14 – Appendix

14.1 Source Code

14.2 Circuit Diagram

14.3 Component Datasheets

14.4 Additional Photographs

Important Instructions

1. All project teams must prepare the report **strictly according to the above format**.
2. The report must contain the **actual hardware implementation details** of the project.
3. The **Block Diagram, Circuit Diagram, Flowchart, and Hardware Model photographs** must be clearly included.
4. Component specifications and relevant datasheets should be provided wherever applicable.
5. The **Embedded C/Python source code** must be included in the Appendix.
6. Testing must include appropriate **test cases, observations, results, and performance analysis**.
7. All figures, tables, diagrams, and photographs must be properly numbered and captioned.
8. References must be provided for research papers, books, websites, datasheets, and other sources used.
9. Students must ensure that the report is **technically correct, properly formatted, and free from plagiarism**.
10. Each project team must be prepared to **demonstrate the working hardware model** during the project review.

All students are instructed to follow the above guidelines without deviation while preparing and submitting the Embedded Hardware Project Report.

**Project Coordinator
Department of Computer Science and Engineering**